

Problem Statement Proposal

SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

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SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

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Description (Background, Detailed Description and Expected Solution)

Background

Remote-sensing imagery is widely used for agricultural monitoring, disaster management, urban planning, forest monitoring, water-resource assessment, infrastructure mapping, and environmental analysis. However, most existing remote-sensing AI solutions are developed as isolated applications for a single predefined task, such as land-cover classification, object detection, visual question answering, or change detection. These systems often require users to understand satellite-data characteristics, GIS workflows, model selection, and task-specific parameters. Consequently, non-expert users may find it difficult to obtain meaningful information from satellite imagery through simple natural-language queries.

Many operational remote-sensing questions cannot always be answered reliably using a single optical image. Relevant information may be distributed across paired or multiple observations acquired at different times or by different sensors. Optical and multispectral imagery provides spectral and contextual information, whereas synthetic aperture radar (SAR) provides complementary structural information and supports day-and-night acquisition through cloud cover. Multitemporal image pairs are required to identify and interpret changes over time, while co-registered optical–SAR pairs can provide more complete and reliable information than either modality alone.

A general-purpose large language model (LLM) or vision-language model (VLM) cannot be expected to perform these specialised tasks reliably without adaptation to remote-sensing imagery, sensor characteristics, and domain-specific terminology. The proposed solution must therefore include remote-sensing fine-tuning or domain adaptation and may employ multiple specialised models for different tasks. BigEarthNet.txt will serve as the primary dataset for adapting image–text representations to multisensor remote-sensing data. VRSBench and RSVQA will be used to evaluate single-image captioning, grounding, and visual question answering, while CDVQA will be used to evaluate multitemporal change-based visual question answering.

The novelty of SatQuery AI lies in its agentic, query-driven framework. Instead of applying a single generic VLM, the system selects and executes suitable remote-sensing specialist models, validates inputs, combines their outputs, and returns an evidence-grounded response.

Detailed Description

The objective is to develop SatQuery AI, a software-based agentic vision-language assistant for analysing single and paired remote-sensing images through natural-language queries. Single-image understanding is a mandatory baseline, while the principal focus is joint reasoning over paired cross-modal and multitemporal imagery.

Defined Input Scope

Single image: One optical/multispectral or SAR image for captioning, visual question answering, and text-guided region grounding.

Cross-modal pair: Co-registered optical/multispectral and SAR images of the same geographic area for joint information extraction and cross-modal analysis.

Bi-temporal pair: Two spatially corresponding images of the same geographic area acquired at different times for change detection, change description, and change-based visual question answering.

Supported formats: GeoTIFF or TIFF for geospatial imagery. PNG and JPEG inputs may be accepted only for the prescribed public benchmark datasets.

Mandatory Functional Scope

- Remote-sensing adaptation: At least one visual or vision-language component must be fine-tuned or otherwise adapted using BigEarthNet.txt or the any open source training data.
- Single-image baseline: Visual question answering shall be mandatory. Each solution must additionally implement either captioning/scene description or text-guided region grounding.

- Multi-image change analysis: Change description or change-based visual question answering from a bi-temporal image pair shall be mandatory. A spatial change map may also be generated where reference masks are available.
- Cross-modal pair analysis: The system must extract complementary information from a co-registered optical/multispectral and SAR image pair.
- Agentic orchestration: The system must automatically select, sequence, and execute the appropriate specialist models or tools according to the query and input configuration.

Representative Queries

- “Describe the land-cover and major objects visible in this image.”
- “Highlight the water body referred to in the query.”
- “What changed between these two dates, and where did the change occur?”
- “Use the optical and SAR images together to identify built-up and water-covered regions.”
- “Has the built-up area increased, decreased, or remained unchanged?”

Agentic Model and Tool Orchestration

The system may use multiple specialised components, such as a remote-sensing VQA or captioning model, a grounding model, a change-understanding or change-VQA model, and an optical–SAR fusion or information-extraction model.

- interpret the query and classify the requested task;
- check the number, modality, format, metadata, and compatibility of the input images;
- select one or more models or tools from a predefined registry;
- configure only permitted task parameters and execute the selected workflow;
- combine textual and spatial outputs, estimate confidence, and return visual evidence; and
- provide an auditable execution summary containing the selected task, model/tool names, and key parameters.

The controller may perform internal task planning; however, only the observable execution trace, including the selected task, models or tools, permitted parameters, and outputs will be evaluated. Internal reasoning text is neither required nor evaluated.

Expected Solution

The expected solution is an interactive GUI or web application with an agentic remote-sensing AI backend. It should accept supported image inputs and natural-language queries, select the appropriate specialist workflow, and return evidence-grounded textual and visual results.

The solution should include:

Input upload and compatibility checking.

A remote-sensing-adapted vision-language component.

Specialist tools for VQA, captioning or grounding, change understanding, and optical–SAR analysis.

An agentic controller for task routing, tool execution, and output integration.

Visual evidence, confidence information, execution summaries, and downloadable reports.

Each solution must demonstrate single-image VQA, one additional single-image task, multitemporal change understanding, optical–SAR paired-image analysis, and agentic model/tool orchestration. A generic LLM or VLM without remote-sensing adaptation will not satisfy the requirements.

Deliverables

An interactive GUI or web application with an agentic remote-sensing AI backend, Codes and models including test and demonstration.

Implementation Scope

The system shall support single optical/multispectral or SAR images, co-registered optical–SAR pairs, and bi-temporal pairs in GeoTIFF/TIFF or approved benchmark formats. It must perform single-image VQA, one additional single-image task, change analysis, optical–SAR joint analysis, and agentic model/tool selection through an interactive GUI or web application.

Evaluation/Judging Criteria

Final evaluation will use prescribed public benchmark test subsets and an ISRO/SAC evaluation dataset. Scores will be normalised before combining different metrics.

Evaluation Component	Metric / Evidence	Weight
Agentic task routing and tool orchestration	Correct task and specialist-model/tool selection, valid execution sequence, input validation, permitted-parameter configuration, and invalid-call rate	20%

Evaluation Component	Metric / Evidence	Weight
Single-image remote-sensing VLM performance	VRSBench and RSVQA VQA accuracy; VRSBench captioning or grounding metrics, depending on the implemented task	20%
Multitemporal change understanding	CDVQA average and overall answer accuracy; change-map F1-score and IoU only where reference change masks are provided	25%
ISRO cross-modal paired-image analysis	Cartosat-2S optical and RISAT SAR pairs: query-answer accuracy, information-extraction F1/IoU, and evidence localisation	25%
Robustness, usability, and reporting	GeoTIFF/TIFF handling, graceful error messages, latency, confidence reporting, visual evidence, and downloadable report	10%

Public benchmarks will be evaluated using the prescribed test splits. The ISRO/SAC evaluation set will contain pre-georeferenced and co-registered Cartosat-2S optical and RISAT SAR image pairs, with task-specific reference answers, labels, bounding boxes, or masks, as applicable. Evaluation annotations will not be disclosed to participating teams.

Dataset Link

Training / Fine-Tuning Dataset

BigEarthNet.txt — primary dataset for remote-sensing adaptation using co-registered Sentinel-1 SAR, Sentinel-2 multispectral imagery, and diverse text annotations. Link: <https://arxiv.org/abs/2603.29630>. **All datasets are available online open source.**

Public Evaluation Benchmarks

- VRSBench — for remote-sensing image captioning, visual grounding, and VQA. Link: <https://arxiv.org/abs/2406.12384>
- RSVQA — for single-image remote-sensing visual question answering. Link: <https://arxiv.org/abs/2003.07333>
- CDVQA — for change-detection-based visual question answering using multitemporal image pairs. Link: <https://arxiv.org/abs/2112.06343>

ISRO/SAC Evaluation Data

A hidden evaluation set of pre-georeferenced and co-registered Cartosat-2S optical imagery and RISAT SAR imagery will be used for cross-modal paired-image analysis. Bi-temporal pairs and corresponding annotations or question-answer references will be provided where required for change-analysis evaluation. **ISRO datasets will be given during testing on hackathon day.**

Category

Software

Theme

Space Technology

Organization (Ministry/Department Name): Space Applications Centre, ISRO, Department of Space

Domain Experts:

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